WORKED SOLUTION

STEP III 2014, Question 10

Mechanics • Hooke's Law

STEP III 2014, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q10

This question has a full guided treatment in *Beyond STEP*, Section 11.2, *Normal Modes*: https://beyondstep.keithmcnulty.org/ch11/normal-modes.html

The key idea

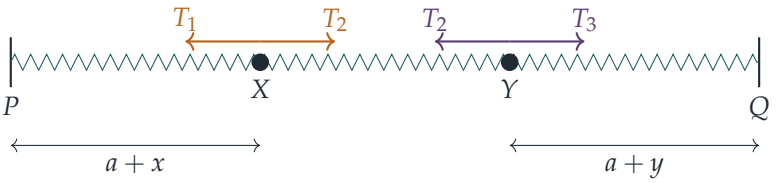
Put everything on one line with a single coordinate, work out the extension of each spring in terms of x and y, and Newton's second law gives two coupled equations. Coupled equations with this symmetry separate when you add and subtract them: $x - y$ and $x + y$ each obey their own simple harmonic equation, with different frequencies. Whether Y ever comes back then turns on whether those two frequencies have a rational ratio.

Solution

The equations of motion

Model X and Y as particles and the springs as light, obeying Hooke's law; the table is smooth, so the only horizontal forces are the spring tensions. Every force acts along the line PQ, and the motion stays on it.

Measure positions along the line from P, in the direction of Q, so P is at 0 and Q at $3a$. Then X is at $a + x$ and Y is at $3a - (a + y) = 2a - y$.



The three springs have lengths and tensions (by Hooke's law, tension $= \lambda \times$ extension$/a$,

negative meaning thrust):

$$PX:\ \text{length } a+x, \qquad T_1 = \frac{\lambda x}{a},$$

$$XY:\ \text{length } (2a-y)-(a+x) = a-x-y, \qquad T_2 = -\frac{\lambda(x+y)}{a},$$

$$YQ:\ \text{length } a+y, \qquad T_3 = \frac{\lambda y}{a}.$$

A spring in tension pulls each end towards the other. So X feels T_1 towards P and T_2 towards Y; taking the direction $P \to Q$ as positive,

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -T_1 + T_2 = -\frac{\lambda}{a}x - \frac{\lambda}{a}(x+y) = -\frac{\lambda}{a}(2x+y),$$

as required. (The signs take care of themselves: if $T_2 < 0$ the spring XY is compressed and pushes X towards P.)

For Y, its position is $2a-y$, so its acceleration in the positive direction is $-\ddot{y}$. It feels T_2 towards X (negative direction) and T_3 towards Q (positive direction):

$$-m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -T_2 + T_3 = \frac{\lambda}{a}(x+y) + \frac{\lambda}{a}y, \qquad \text{so} \qquad m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -\frac{\lambda}{a}(x+2y).$$

This is the first equation with x and y swapped, as the symmetry of the set-up (reflect in the midpoint of PQ) says it must be.

The combination $x-y$

Subtracting the two equations,

$$m\frac{\mathrm{d}^2}{\mathrm{d}t^2}(x-y) = -\frac{\lambda}{a}\left[(2x+y)-(x+2y)\right] = -\frac{\lambda}{a}(x-y).$$

With $\omega^2 = \lambda/(ma)$ this is $\dfrac{\mathrm{d}^2}{\mathrm{d}t^2}(x-y) = -\omega^2(x-y)$, simple harmonic motion, so

$$x - y = A\cos\omega t + B\sin\omega t.$$

Initially $XP = a$ and $YQ = \frac{1}{2}a$, so $x = 0$ and $y = -\frac{1}{2}a$ at $t = 0$, giving $x - y = \frac{1}{2}a$ and hence $A = \frac{1}{2}a$. The particles are released from rest, so $\dot{x} = \dot{y} = 0$ at $t = 0$; differentiating, $\omega B = 0$, so $B = 0$. Thus

$$x - y = \tfrac{1}{2}a\cos\omega t.$$

The combination $x+y$

Adding the equations instead,

$$m\frac{\mathrm{d}^2}{\mathrm{d}t^2}(x+y) = -\frac{\lambda}{a}(3x+3y), \qquad \text{so} \qquad \frac{\mathrm{d}^2}{\mathrm{d}t^2}(x+y) = -3\omega^2(x+y).$$

This is simple harmonic motion with angular frequency $\sqrt{3}\,\omega$, so $x + y = C\cos\sqrt{3}\,\omega t + D\sin\sqrt{3}\,\omega t$. At $t = 0$, $x + y = -\frac{1}{2}a$ and its derivative is 0, so $C = -\frac{1}{2}a$ and $D = 0$:

$$x + y = -\tfrac{1}{2}a\cos\sqrt{3}\,\omega t.$$

Y never returns

Half the difference of the two results gives

$$y = \tfrac{1}{2}\big[(x+y) - (x-y)\big] = -\tfrac{1}{4}a\big(\cos\omega t + \cos\sqrt{3}\,\omega t\big).$$

(Similarly $x = \tfrac{1}{4}a(\cos\omega t - \cos\sqrt{3}\,\omega t)$.) Y is back at its initial position exactly when $y = -\tfrac{1}{2}a$, that is, when

$$\cos\omega t + \cos\sqrt{3}\,\omega t = 2.$$

? **Ask yourself** Both cosines are periodic, so surely they line up again eventually?

Each cosine is at most 1, so the sum equals 2 only if both equal 1 at the same time $t > 0$. That needs

$$\omega t = 2\pi n \quad \text{and} \quad \sqrt{3}\,\omega t = 2\pi k$$

for integers n, k, with $n \geq 1$ because $t > 0$. Dividing, $\sqrt{3} = k/n$ would be rational, which it is not. So there is no such $t > 0$, and Y never returns to its initial position. The two periods $2\pi/\omega$ and $2\pi/(\sqrt{3}\,\omega)$ are incommensurable, so the two oscillations never get back in step, although they come arbitrarily close to doing so.

Checks

- At $t = 0$ the formulae give $x = \tfrac{1}{4}a(1-1) = 0$ and $y = -\tfrac{1}{4}a(1+1) = -\tfrac{1}{2}a$, matching the starting positions, and both derivatives vanish there.
- The length of spring XY is $a - x - y = a + \tfrac{1}{2}a\cos\sqrt{3}\,\omega t$, which stays between $\tfrac{1}{2}a$ and $\tfrac{3}{2}a$, so the particles never collide; likewise $PX = a + x \geq \tfrac{1}{2}a$ and $YQ = a + y \geq \tfrac{1}{2}a$. The model stays physically sensible throughout.
- Stepping the motion numerically with $m = a = 1$, $\lambda = 2$ reproduces the formulae for x and y to better than 10^{-10} up to $t = 20$. A computer search over $0 < \omega t \leq 20\,000$ finds that $\cos\omega t + \cos\sqrt{3}\,\omega t$ gets as close as 1.999995 to 2, but never reaches it.

Common slips

- Getting the sign of the Y equation wrong: y is measured from Q, so increasing y moves Y in the *negative* direction.
- Writing the length of XY as $a + x + y$ instead of $a - x - y$; both x and y increasing make the middle spring shorter.
- Arguing only that the two periods differ. Different periods alone are not enough (periods 1 and 2 realign at $t = 2$); what matters is that their ratio $\sqrt{3}$ is irrational.

Beyond STEP

The combinations $x - y$ and $x + y$ are the *normal modes* of this system: the eigenvectors of the matrix $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, with eigenvalues 1 and 3 giving the squared frequencies ω^2 and $3\omega^2$. Every small oscillation of a coupled system splits into normal modes like this, and the motion is periodic only when all the frequency ratios are rational; otherwise it is quasi-periodic, never repeating but returning arbitrarily close to its start. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers coupled oscillators, and Strang, *Introduction to Linear Algebra*, explains the eigenvector method behind them.

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